

Physics Homework: Kinematics, Forces, and Energy

Instructions

Show all work. Follow each step carefully. Focus on **process, not just answers**.

1 1. Kinematics (Guided)

Problem 1

A car starts from rest and accelerates at 2 m/s^2 for 6 seconds.

1. What are the known values? (v_0 , a , t)

2. Which equation will you use?

3. Find the final velocity.

4. Now find the displacement.

Problem 2

An object has initial velocity 10 m/s and slows down at 2 m/s^2 .

1. What is the acceleration sign?
2. How long does it take to stop?
3. How far does it travel before stopping?

2 2. Forces (Guided)

Problem 3

A 5 kg box is pushed across a horizontal surface with $\mu_k = 0.2$.

1. Draw a free body diagram (list forces below).
2. What is the normal force?
3. Find the friction force.

4. What is the acceleration?

Problem 4 (Incline)

A 10 kg block slides down a 30° incline (no friction).

1. Draw the free body diagram.
2. Break weight into components.
3. What force causes motion?
4. Find acceleration.

3 3. Energy (Guided)

Problem 5

A 2 kg object is dropped from a height of 10 m.

1. What type of energy does it start with?

2. Write the energy equation in symbols.

3. Replace each term with expressions.

4. Solve for final speed.

Problem 6 (Friction)

A 3 kg block slides 4 m across a surface with $\mu_k = 0.25$.

1. What type of force is friction (conservative or non-conservative)?

2. Find the friction force.

3. Find the work done by friction.

4. Is the work positive or negative? Why?

Problem 7 (Energy + Motion)

A block starts from rest at height 5 m and slides down without friction.

1. What energy does it start with?
2. What energy does it end with?
3. Write the energy equation.
4. Solve for final speed.

4 4. Mixed Challenge (Step-by-Step)

Problem 8

A 2 kg block starts with speed 6 m/s and slides across a rough surface ($\mu_k = 0.25$) until it stops.

1. What is the initial energy?

2. What is the final energy?
3. What role does friction play?
4. Write the full energy equation.
5. Solve for distance traveled.

Reflection

- When should you use kinematics vs energy?
- What does negative work done on an object mean physically?